

PRAMANA

Evidence-Centric Attribution Platform for Dark Web Threat Actors — SIH 2026, Problem Statement 26151 (NTRO)

AI/ML Workstream: Module Ownership & Dataset Mapping (v2 — aligned to Master Blueprint)

This revision replaces the earlier generic entity-resolution plan with PRAMANA's actual architecture: the evidence ledger, the nine evidence families (F1-F9), capped independence-aware log-LR fusion, counter-evidence/must-not-link vetoes, and the Cllr/Tippett evaluation harness, as defined in the Master Blueprint. The AI/ML workstream is split at the same boundary the Blueprint itself uses: deterministic/ML **evidence generation** versus **resolution, fusion and validation**. Every module below states which of your forwarded datasets feeds it, and what ledger object it actually produces — not a generic API response.

Team split	Person A - List A: Evidence Generation (extraction, canonicalisation, rarity, per-family scoring) Person B - List B: Resolution & Fusion (entity resolution, counter-evidence, log-LR fusion, calibration, evaluation harness)
Non-negotiable rule (Blueprint, hard)	No LLM, embedding, or ML score ever writes directly to evidence, pair_score, assessment, or must_not_link. Deterministic extractors are authoritative; ML/LLM outputs are advisory or must pass a validator before promotion.
Deployment target	Prototype stays on a single PostgreSQL 16 instance per the Blueprint (no Neo4j/Elasticsearch/Kafka). ML inference services (extractors, scorers, embeddings) packaged as small FastAPI containers, deployable on Hugging Face Spaces/Inference Endpoints or Cloudflare Workers AI, called synchronously by the FastAPI backend — not by the frontend directly.
Rejected by Blueprint (do not build)	Dempster-Shafer/Subjective Logic fusion, GNN-based linkage (prototype scale), autonomous agentic attribution, real-world identity output, percentage-match UI, Monero tracing, passive onion enumeration.

List A — Evidence Generation (Extraction, Canonicalisation, Per-Family Scoring)

Owner: Person A. Meta-goal: turn sealed, normalised documents into typed **evidence** rows — each tagged with an evidence family (F1-F9), a raw log-LR contribution, a rarity factor, and an independence key. This person does not decide attribution, rank candidates, or fuse scores — those are List B. Every module here maps to a pipeline stage in the Blueprint's lifecycle: EXTRACTION -> CANONICALIZATION -> RARITY INDEXING -> EVIDENCE CREATION.

Module	What to build	Dataset(s) used	Output / API result
Deterministic extractors (F1/F2/F3/F4)	Regex/checksum validators for BTC/XMR addresses, PGP blocks, Jabber/Session/Telegram/email identifiers, onion addresses; parses raw text into typed indicator mentions. Authoritative per Blueprint — never ML-generated.	Darknet Market Archives (Gwern, 2013-2015) for vendor-profile text containing PGP blocks and contacts; Cambridge Cybercrime Centre corpora for forum-post text.	indicator rows (type, value, doc_id, extractor_version) written directly to the ledger.
PGP key-metadata forensics (F1)	Parse key creation timestamp, algorithm/bit-size, cipher-preference list, GnuPG version artefact, subkey structure, UID conventions — links *different* keys of one operator, survives text laundering.	Darknet Market Archives (Gwern) vendor PGP key blocks; synthetic key-metadata cases planted in RANGE-SIM for validation.	pgp_key rows + F1 evidence with raw_log_lr, capped at 4.0 per Blueprint family table.
Wallet clustering (F2, capped)	Co-spend/CIOH heuristics for address clustering; CoinJoin/PayJoin/exchange-batching detectors that suppress rather than encourage merges, per Blueprint ("a shared address never triggers a merge").	BitcoinHeist (UCI ML Repo) for labeled ransomware wallet behaviour patterns; Stanford SNAP for transaction-graph structure/topology reference.	wallet_address.cluster_id + F2 evidence, cap 2.0 log-LR, coinjoin_flag/exchange_flag set where detected.
Infrastructure fingerprinting (F3)	TLS/SAN parsing, favicon/asset/header hashing, clock-skew estimation, OnionBalance instance-count checks, mirror/Onion-Location consistency — targeted probing of known addresses only, never enumeration.	Darknet Market Archives site snapshots; certificate-transparency log data (public CT logs) for TLS/SAN cross-referencing; RANGE-TOR (self-owned Tor testbed) for controlled validation.	infra_artefact rows + F3 evidence with mandatory shared_host_flag before any origin claim.
Canonicalisation & clone detection (independence anchor)	SHA-256 exact match; SimHash (64-bit, Hamming <=3) for near-duplicate text; perceptual hashing (pHash/dHash) for images; shingle containment for quoted text; clone/payment-address-substitution detector for phishing clones.	Darknet Market Archives text + product images; ImageNet used only as a pretrained-backbone source for the pHash/feature extractor (not domain data).	canonical_artefact_id + doc_relation edges (mirrors/quotes/clone_of) — this is what makes the independence_key possible downstream.
Rarity indexing (feeds every family)	Per-indicator account/site frequency count -> IDF-like rarity score; hub flag when occurrence count exceeds threshold tau (Blueprint default tau=12).	Computed directly from the ingested corpus (Cambridge Cybercrime Centre + Darknet Market Archives), snapshot-versioned per Blueprint ("rarity recomputed per snapshot").	indicator.rarity, indicator.is_hub fields; hub-flagged indicators forced to zero contribution.

Module	What to build	Dataset(s) used	Output / API result
Language/script ID & NER	Language + Unicode script detection (catches homoglyph/mixed-script evasion); multilingual NER for handles, products, org/location mentions, embedded crypto addresses in free text.	Cambridge Cybercrime Centre + Darknet Market Archives text; DarkBERT as the base language model since it is pretrained specifically on dark-web text; s2w-ai/CoDA (HF) as supplementary fine-tune/eval data.	document.lang/script fields; NER output staged as machine_suggested indicators, requiring validator pass before promotion (Blueprint hard rule).
Template/artefact fingerprinting (F5)	Structural parsing of listing/post templates — section order, table layout, unit and price-rounding conventions, signature-block form — the direct, laundering-resistant answer to LLM-rewritten prose.	Darknet Market Archives listings/posts across multiple markets (needed specifically to compare template structure cross-market).	document.template_hash + F5 evidence, cap 2.5 log-LR.
Gated stylometry & laundering detection (F6)	Function-word frequencies, character n-grams, style embeddings for weak corroboration only; a laundering detector flags LLM-mediated/translated text and marks it stylistically inadmissible instead of silently scoring it.	Extended Brennan-Greenstein Stylometry Corpus for ground-truth authorship features; PAN shared-task datasets (Authorship Verification track) for validation methodology and laundering-robustness benchmarks.	F6 evidence, hard-capped at 1.0 log-LR; laundering_flag on affected documents (never silently fused).
Style-discontinuity detection (inverse stylometry)	Sliding-window drift detection on the *same* account over time — flags handover/compromise/staff -rotation rather than cross-account similarity. The Blueprint's key innovation the dossier missed.	PAN authorship-verification datasets for the drift-detection methodology; applied to Cambridge/Darknet Market Archive per-account timelines.	style_shift_flag + confidence, used as F6 corroboration and as a standalone handover signal (not a merge trigger).
Behavioural/temporal detectors (F7)	Migration-window co-timing detector around lifecycle shocks (market shutdowns, takedowns); catalogue/price-structure continuity; explicitly excludes timezone-from-posting-rhythm (Blueprint marks this obsolete).	Cambridge Cybercrime Centre + Darknet Market Archives timestamped data, especially known migration events (e.g. Silk Road -> Silk Road 2).	F7 evidence, cap 0.7 log-LR, event-anchored rather than raw-timestamp-anchored.

End result of List A: populated evidence rows (family, raw_log_lr, rarity_factor, independence_key, polarity) plus canonical_artefact/doc_relation records. This is the exact contract List B consumes — no similarity scores, no rankings, no fused numbers.

List B — Entity Resolution, Fusion & Validation

Owner: Person B. Meta-goal: consume List A's evidence rows and turn them into a calibrated, contestable **assessment** object — never a raw score. Covers pipeline stages CANDIDATE GENERATION -> ENTITY RESOLUTION -> COUNTER-EVIDENCE -> INDEPENDENCE GROUPING -> FUSION -> ASSESSMENT, plus the evaluation harness that makes the whole design falsifiable.

Module	What to build	Dataset(s) used	Output / API result
Candidate generation (blocking)	Multi-key blocking on identifiers, pHash buckets, template hash, username-variant embeddings, ANN search to avoid $O(n^2)$ comparison while keeping recall measurable.	Recall measured against RANGE-SIM's hidden ground truth (planted multi-account operators); applied over Cambridge Cybercrime Centre + Darknet Market Archive accounts.	candidate_pair rows with blocking_key, feeding pairwise scoring.
Entity resolution (Fellegi-Sunter / Splink)	Probabilistic record linkage using Splink as reference implementation; m/u parameter estimation by EM; per-pair evidence score; constrained clustering with must-not-link vetoes, component-size caps, and hub removal — explicitly no automatic transitive closure.	Trained/validated on RANGE-SIM synthetic ground truth (planted operators, decoy pairs sharing misleading single-family signals); cross-checked against Darknet Market Archives' known vendor migrations as real positive pairs.	pair_score rows + provisional cluster assignments; giant-component and purity alarms if a cluster grows abnormally.
Counter-evidence engine	Five contradiction-class detectors: temporal impossibility, contradictory identifiers, geolocation/language conflict, impersonation/compromise patterns, shared-ecosystem-not-shared-control explanations. Hard contradictions veto to EXCLUDED (no numeric score); soft contradictions subtract from the balance.	Rules validated against RANGE-SIM's planted decoy pairs (deliberately misleading single-family matches) and RANGE-TOR's clean controls (to measure false positives).	evidence(polarity=neg) rows + must_not_link constraints (hard/soft), consumed before fusion.
Independence grouping & within-family damping	Group evidence by (canonical_artefact_id, family, source_independence_cluster); strongest item per group counts fully, later independent groups damped by $\lambda=0.2$ (Blueprint default), subject to per-family caps (F1=4.0, F2=2.0, F3=2.5, F4=3.0, F5=2.5, F6=1.0, F7=0.7, F8=1.0, F9=3.0).	Damping parameter (λ) and caps tuned against RANGE-SIM's mirror-heavy planted cases via with-vs-without-grouping ablation.	evidence_group rows with damped_log_lr + discount_reason, forming the visible 'independence discount' in the Evidence Balance Sheet.
Log-LR fusion & assessment	Capped, damped log10-LR sum across families minus counter-evidence, global ceiling 4.0, minimum $k \geq 2$ independent families required, hard must-not-link short-circuits to EXCLUDED.	Calibration inputs from the same RANGE-SIM/RANGE-TOR labelled pairs; no live/real-marketplace data used, per Blueprint constraints.	assessment object: log_lr, calibrated_posterior, verbal_band, family_count_k, defence_hypothesis, limitations[].
Calibration	Logistic/isotonic calibration mapping raw fused log-LR to a stated posterior, reported against an explicitly declared reference population (never population-independent probability).	Fit and cross-validated on RANGE-SIM ground truth with temporal train/test splits (train on earlier synthetic months, test on later) to avoid leakage.	params_version-pinned calibration map + reliability diagram, shown as a first-class product surface.

Module	What to build	Dataset(s) used	Output / API result
Evaluation harness (Cllr / Tippett / clustering metrics)	Cllr and Cllr_min for LR calibration/discrimination quality; Tippett plots for same-vs-different-actor separation; B-cubed and Variation of Information for cluster correctness; false-merge rate as the headline safety metric; adversarial-degradation ablations (mirror flooding, hub creation, laundering).	RANGE-SIM (seeded synthetic multi-platform environment, hidden ground truth kept in an isolated schema) and RANGE-TOR (self-owned Tor testbed with known signal classes) — both purpose-built controlled test environments, not the public datasets.	Metrics report + plots surfaced directly in the analyst UI (Evaluation Science screen), not buried in an appendix.
LLM copilot (read-only, evidence-grounded)	DarkBERT / general LLM strictly for evidence-grounded Q&A, query expansion, and drafting assistance for analyst reports. Every answer must cite evidence IDs; uncited answers are not rendered; no write path to core tables.	DarkBERT (pretrained on dark-web text) as the base model, retrieving only from already-computed evidence/assessment records.	Human-readable, evidence-cited explanation text — never a match/no-match verdict.

End result of List B: a reviewed assessment — e.g. “STRONG band, log10 LR 3.1, k=3 independent families, defence hypothesis: shared hosting provider, no hard contradictions” — exactly the object rendered in the Evidence Balance Sheet, never a bare percentage.

Dataset Reference Summary

Mapping of every source you shared to the evidence family / module it actually supports under the Blueprint's taxonomy. Per the Blueprint's own citation-quality rule, none of these are used as a stand-in for RANGE-SIM/RANGE-TOR ground truth in evaluation — they are extraction/training data and, where noted, real-world corroborating cases only.

Dataset / Source	Type	Evidence family / module it supports
Cambridge Cybercrime Centre datasets	Forum/marketplace crawl corpora, timestamps, metadata	F3 infra, F4 contact, F7 behavioural/temporal, language/NER, template fingerprinting (F5), extractor training/eval
BitcoinHeist (UCI ML Repo)	Labelled Bitcoin ransomware wallet dataset	F2 financial - wallet clustering heuristics, CoinJoin/exchange-batching detector training
Stanford Large Network Dataset Collection (SNAP)	Graph/network datasets	Reference topology for wallet transaction-graph structure and for tuning the independence-grouping / hub-suppression logic
Darknet Market Archives, Gwern.net (2013-2015)	Scraped market listings, vendor profiles, PGP keys, product images	F1 PGP metadata, F4 contact, F5 template & content artefacts, canonicalisation/clone detection, NER, stylometry (F6), known-migration positive pairs for entity resolution
ImageNet	General-purpose labelled image dataset	Pretrained CNN backbone only, for the perceptual-hash/feature extractor used in F5 duplicate-image detection - not domain data, per Blueprint's rejection of using generic data as ground truth
s2w-ai/CoDA (Hugging Face)	Dark-web related dataset (HF hub)	Supplementary fine-tune/eval data for NER and language/script-ID modules
Extended Brennan-Greenstein Stylometry Corpus	Authorship/stylometry ground truth	F6 gated stylometry feature extraction, style-discontinuity (inverse stylometry) detector
PAN (shared tasks)	Authorship verification/attribution benchmark datasets	Stylometry validation methodology and laundering-robustness benchmarking - informs F6 caps, not a replacement for RANGE-SIM
DarkBERT	Pretrained language model for dark-web text	Base LM for NER/language modules on dark-web text; base model for the read-only, evidence-grounded LLM copilot

Not covered by your forwarded links, but required by the Blueprint and must be built/sourced separately: RANGE-SIM (seeded synthetic environment with hidden ground truth, isolated answer-key schema) and RANGE-TOR (self-owned Tor testbed for infrastructure fingerprinting validation). These are the actual evaluation ground truth — the public datasets above are for extraction/training only.

Deployment Plan (Prototype)

- Keep the system-of-record on a single PostgreSQL 16 instance per the Blueprint (ledger, evidence, pair_score, assessment tables) - do not split this across services.
- Package each ML/extraction module (PGP metadata parser, pHash/SimHash canonicaliser, NER, stylometry, entity-resolution scorer) as a small FastAPI container, callable by the backend FastAPI service - never called directly by the frontend.
- Host the heavier inference models (stylometry embeddings, NER, DarkBERT-based copilot) on Hugging Face Spaces/Inference Endpoints; keep lightweight deterministic logic (rarity lookup, fusion arithmetic, calibration map) close to the Postgres backend or on Cloudflare Workers for low latency.

- Expose the Evidence Balance Sheet's data (evidence list, family, raw/damped LR, counter-evidence, fused assessment) as one coherent read endpoint so the frontend renders exactly the ledger's own objects, never a re-derived score.
- Surface the evaluation harness (Cllr, Tippett, reliability diagram, false-merge rate) as its own product screen/endpoint, computed against RANGE-SIM/RANGE-TOR - fed live during the demo, not a static appendix slide.

Handoff contract: before coding starts, both team members should freeze the evidence-row schema (family, raw_log_lr, rarity_factor, independence_key, polarity, validity_interval) exactly as defined in the Blueprint's data model (Section 7-8) - this is the only interface that can silently break the split between List A and List B.

Claim Discipline Reminder

Per the Blueprint's own checklist: say "design target $\leq$ 2% false-merge rate," never "we achieve 2% false merges" before RANGE-SIM benchmarking is run. Say "calibrated investigative lead," never "proves who the actor is." Say "LLM assistance is read-only and evidence-grounded," never "our AI autonomously attributes actors." This applies to how this AI/ML module list itself gets presented to judges - the modules above are designed and specified; none are claimed as benchmarked yet.